The Uribe–Yue Small Open Economy Model: Derivation and Dynare Translation

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Overview

This note derives, equation by equation, the small open economy model of Uribe and Yue, "Country Spreads and Emerging Countries: Who Drives Whom?" (reproduced as Chapter 6 of Schmitt-Grohé and Uribe, *Open Economy Macroeconomics*), further translating it into non-linear Dynare equation for each block.

1 Technology

Derivation

Output is produced with a constant-returns Cobb-Douglas technology in capital and labor, with capital share $\alpha \in (0, 1)$:

$$y_t = k_t^{\alpha} h_t^{1-\alpha}.$$

This equation is used throughout the chapter (it is the model's eq. 6.2). It is standard and needed to derive the two firm first-order conditions below.
PDF equation number: (6.2) [Cobb-Douglas technology].

Dynare (non-linear)

model block

```
exp(yy) = exp(k)^ALFA * exp(h)^(1-ALFA);
```

(All quantity variables are carried as logs, exponentiated where a level is needed – this convention is used consistently below.)

2 The firm's problem: working-capital-in-advance

Derivation

The firm rents capital and hires labor to maximize static profits

$$\pi_t = k_t^{\alpha} h_t^{1-\alpha} - w_t h_t \left[1 + \eta \left(\frac{R_t^d - 1}{R_t^d}\right)\right] - u_t k_t,$$

Taking the first-order conditions with respect to h_t and k_t:

$$F_h(k_t, h_t) = w_t \left[1 + \eta \left(\frac{R_t^d - 1}{R_t^d}\right)\right] \qquad \Longrightarrow \qquad w_t = \frac{(1-\alpha) k_t^{\alpha} h_t^{-\alpha}}{1 + \eta \left(R_t^d - 1\right)/R_t^d},$$

$$F_k(k_t, h_t) = u_t \qquad \Longrightarrow \qquad u_t = \alpha k_t^{\alpha-1} h_t^{1-\alpha}.$$

The competitive, constant-returns firm earns zero profits in equilibrium, so $w_t h_t + u_t k_t = y_t$ – a fact used repeatedly below (budget constraint, trade balance identity).
PDF equation numbers: (6.3) for the labor condition, (6.4) for the capital condition. The shadow rate R_t^d used here is defined in eq. (6.19), derived in §8 below (it depends on the household's debt-adjustment cost, so is derived after the household's problem).

Dynare (non-linear)

model-local variables, used throughout

```
#rd = exp(r)/(1-PSSI*(d-DBAR));                              % R^d_t, eq.
    (6.19)
#w  = (1-ALFA)*exp(k)^ALFA*exp(h)^(-ALFA) / (1+ETA*(rd-1)/rd);  % eq.
     (6.3)
```

```
#u   = ALFA*exp(k)^(ALFA-1)*exp(h)^(1-ALFA);                    % eq.
    (6.4)
```

3 Capital accumulation and gestation lags

Derivation

Producing a unit of new capital requires spreading 1/4 unit of investment goods over four consecutive quarters (a "time-to-build" technology). Let s_{it} denote the number of investment projects *started* i quarters ago, $i = 0, 1, 2, 3$. Gross investment in period t is the sum of the per-quarter installments on every vintage currently under construction:

$$i_t = \frac{1}{4} \sum_{i=0}^{3} s_{it}. \tag{6.5}$$

A project started i periods ago becomes, next period, a project started $i + 1$ periods ago:

$$s_{i+1,t+1} = s_{it}, \qquad i = 0, 1, 2. \tag{6.6}$$

Once a project reaches its fourth and final quarter (s_{3t}), it is converted into installed capital subject to the adjustment-cost function Φ derived above:

$$k_{t+1} = (1 - \delta)k_t + k_t\,\Phi\left(\frac{s_{3t}}{k_t}\right). \tag{6.7}$$

PDF equation numbers: (6.5), (6.6), (6.7).

Dynare (non-linear)

model block

```
% eq. (6.5) -- investment aggregator
exp(iv) = (exp(s0)+exp(s1)+exp(s2)+exp(s3))/4;

% eq. (6.6) -- project pipeline (i=0,1,2)
exp(s1) = exp(s0(-1));
exp(s2) = exp(s1(-1));
exp(s3) = exp(s2(-1));

% eq. (6.7) -- capital accumulation with adjustment costs
#ac = exp(s3)/exp(k) - (PHI/2)*(exp(s3)/exp(k)-DELTA)^2;    % Phi(s3/k
    )
exp(k(+1)) = (1-DELTA)*exp(k) + exp(k)*ac;
```

4 Preferences

Derivation

Households have external-habit, CRRA period utility over a composite of consumption net of habit and hours (a Greenwood-Hercowitz-Huffman-type specification, which removes the wealth effect on labor supply and keeps hours easy to characterize in closed form):

$$U(c_t - \mu\tilde{c}_{t-1}, h_t) = \frac{\left(c_t - \mu\tilde{c}_{t-1} - \dfrac{h_t^{\omega}}{\omega}\right)^{1-\gamma} - 1}{1-\gamma}, \tag{6.8}$$

where $\tilde{c}_{t-1}$ is the (taken-as-given) cross-sectional average of consumption in $t-1$, $\mu \in (0,1)$ the intensity of external habit formation, $\gamma > 0$ the curvature parameter, and $\omega > 1$ governs the Frisch elasticity of labor supply $(1/(\omega-1))$. Lifetime utility is

$$E_0 \sum_{t=0}^{\infty} \beta^t U(c_t - \mu\tilde{c}_{t-1}, h_t).$$

PDF equation number: (6.8).

Dynare (non-linear)

U itself never appears as a stand-alone equilibrium condition – only its partial derivatives U_c, U_h do, inside the Euler equations of §7. It is recorded here for reference and reused as a model-local variable there:

model-local variable, reused in §7

```
#composite = exp(c(+1)) - MU*exp(c) - exp(h(+1))^OMEGA/OMEGA;
```

5 Household budget constraint and borrowing limit

Derivation

Households own capital and hold a one-period bond d_t (positive $d_t =$ household debt). Each period's sources of income (wages, capital rentals, firm profits, net of new bond issuance) fund consumption, investment, and debt service, net of a portfolio-adjustment cost $\Psi(d_t)$ on the bond position:

$$d_t = R_{t-1}d_{t-1} + \Psi(d_t) + c_t + i_t - w_t h_t - u_t k_t - \pi_t, \tag{6.9}$$

subject to the no-Ponzi-scheme condition

$$\lim_{j\to\infty} E_t \frac{d_{t+j+1}}{\prod_{s=0}^{j} R_{t+s}} \leq 0. \tag{6.10}$$

Because the representative firm is competitive with constant returns, $\pi_t = 0$ and $w_t h_t + u_t k_t = y_t$ (Euler's theorem), so in equilibrium the budget constraint simplifies to

$$d_t = R_{t-1}d_{t-1} + \Psi(d_t) + c_t + i_t - y_t.$$

PDF equation numbers: (6.9), (6.10).

Dynare (non-linear)

model block

```
exp(r(-1))*d(-1) + (PSSI/2)*(d-DBAR)^2 - w*exp(h) - u*exp(k)
  + exp(c) + exp(iv) = d;
```

6 The household's Lagrangian

Setup

Because production and absorption decisions are constrained to be made one period in advance, in period t the variables c_t, h_t, s_{it} $(i = 0, 1, 2, 3)$ are already predetermined (chosen at $t-1$). The household therefore chooses, at date t: $c_{t+1}, h_{t+1}, s_{i,t+1}$ $(i = 0, 1, 2, 3)$, d_t, and k_{t+1}, to maximize lifetime utility subject to the budget constraint (6.9), the project pipeline and capital-accumulation equations (6.5)–(6.7), and the no-Ponzi condition (6.10). The associated Lagrangian is

$$\begin{aligned}\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \Bigg\{ & U(c_t - \mu\tilde{c}_{t-1}, h_t) \\ & + \lambda_t \Big[d_t - R_{t-1}d_{t-1} - \Psi(d_t) + w_t h_t + u_t k_t + \pi_t - \tfrac{1}{4}\textstyle\sum_{i=0}^{3} s_{it} - c_t \Big] \\ & + \lambda_t q_t \Big[(1-\delta)k_t + k_t \Phi\Big(\tfrac{s_{3t}}{k_t}\Big) - k_{t+1} \Big] \\ & + \lambda_t \textstyle\sum_{i=0}^{2} \nu_{it}(s_{it} - s_{i+1,t+1}) \Bigg\},\end{aligned}$$

where $\beta^t\lambda_t$, $\beta^t\lambda_t q_t$, and $\beta^t\lambda_t\nu_{it}$ are the multipliers on the budget constraint, the capital-accumulation equation, and the project-pipeline equations, respectively. This is the Lagrangian as it appears in the text, immediately following (6.10).

Each first-order condition below is obtained by differentiating $\mathcal{L}$ with respect to one choice variable and setting the result to zero. Because every choice variable dated $t+1$ appears in exactly *two* consecutive terms of the sum (once at its own date $t+1$, once as a state entering the date-t constraint), each derivative collects exactly two terms – one at β^t, one at β^{t+1} – which is why every resulting Euler equation pairs a date-t and a date-$(t+1)$ object.

7 FOC 1: consumption Euler equation

Derivation

Collect every place c_{t+1} appears in $\mathcal{L}$: inside $U(c_{t+1} - \mu\tilde{c}_t, h_{t+1})$ at coefficient β^{t+1}, and inside $-\lambda_{t+1}c_{t+1}$ (from the budget constraint at $t+1$), also at coefficient β^{t+1}. Differentiating and dividing through by β^{t+1}:

$$\frac{\partial \mathcal{L}}{\partial c_{t+1}} = 0 \implies U_c(c_{t+1} - \mu\tilde{c}_t, h_{t+1}) - \lambda_{t+1} = 0 \implies E_t\lambda_{t+1} = E_t\, U_c(c_{t+1} - \mu\tilde{c}_t, h_{t+1}), \tag{6.11}$$

with, from (6.8),

$$U_c(c_{t+1} - \mu\tilde{c}_t, h_{t+1}) = \left(c_{t+1} - \mu\tilde{c}_t - \frac{h_{t+1}^{\omega}}{\omega} \right)^{-\gamma}.$$

PDF equation number: (6.11).

Dynare (non-linear)

model block

```
exp(la(+1)) = (exp(c(+1)) - MU*exp(c) - exp(h(+1))^OMEGA/OMEGA)^(-
    GAMA);
```

8 FOC 2: labor supply

Derivation

Collect every place h_{t+1} appears: inside $U(c_{t+1} - \mu\tilde{c}_t, h_{t+1})$ and inside $+\lambda_{t+1} w_{t+1} h_{t+1}$ (from the budget constraint at $t+1$), both at coefficient β^{t+1}:

$$\frac{\partial \mathcal{L}}{\partial h_{t+1}} = 0 \implies U_h(c_{t+1} - \mu\tilde{c}_t, h_{t+1}) + \lambda_{t+1} w_{t+1} = 0 \implies E_t[w_{t+1}\lambda_{t+1}] = -E_t\, U_h(c_{t+1} - \mu\tilde{c}_t, h_{t+1}), \tag{6.12}$$

with

$$U_h(c_{t+1} - \mu\tilde{c}_t, h_{t+1}) = -\left(c_{t+1} - \mu\tilde{c}_t - \frac{h_{t+1}^{\omega}}{\omega}\right)^{-\gamma} h_{t+1}^{\omega-1}.$$

PDF equation number: (6.12).

Dynare (non-linear)

model block

```
#rd_lead = exp(r(+1))/(1-PSSI*(d(+1)-DBAR));
#w_lead  = (1-ALFA)*exp(k(+1))^ALFA*exp(h(+1))^(-ALFA) / (1+ETA*(
    rd_lead-1)/rd_lead);
(exp(c(+1))-MU*exp(c)-exp(h(+1))^OMEGA/OMEGA)^(-GAMA)*exp(h(+1))^(
    OMEGA-1)
  = exp(la(+1))*w_lead;
```

9 FOC 3: bond Euler equation

Derivation

d_t appears twice: inside $\lambda_t[d_t - \Psi(d_t)]$ at β^t, and inside $-\lambda_{t+1} R_t d_t$ (the $-R_t d_t$ term of the budget constraint one period later, at β^{t+1}):

$$\frac{\partial \mathcal{L}}{\partial d_t} = 0 \implies \beta^t \lambda_t\big[1 - \Psi'(d_t)\big] - \beta^{t+1} E_t[\lambda_{t+1} R_t] = 0 \implies \lambda_t\big[1 - \Psi'(d_t)\big] = \beta R_t\, E_t \lambda_{t+1}. \tag{6.13}$$

Interpretation: because of the cost of adjusting the bond position, the relevant rate of return households equate to their intertemporal marginal rate of substitution is not the market rate R_t but the shadow rate $R_t^d \equiv R_t/(1 - \Psi'(d_t))$ – see (6.19), §8.
PDF equation number: (6.13).

Dynare (non-linear)

```
model block
exp(la)*(1-PSSI*(d-DBAR)) = BETTA*exp(r)*exp(la(+1));
```

10 FOC 4–6: pricing the investment-project pipeline

Derivation: $s_{0,t+1}$

$s_{0,t+1}$ appears in $-\lambda_{t+1}\frac{1}{4}s_{0,t+1}$ (investment aggregator inside the budget constraint, at $t+1$) and in $+\lambda_{t+1}\nu_{0,t+1}s_{0,t+1}$ (the $i=0$ term of the pipeline multiplier sum, also at $t+1$), both at coefficient β^{t+1}:

$$\frac{\partial \mathcal{L}}{\partial s_{0,t+1}} = 0 \implies E_t\lambda_{t+1}\nu_{0,t+1} = \tfrac{1}{4}E_t\lambda_{t+1}. \tag{6.14}$$

Derivation: $s_{1,t+1}$

$s_{1,t+1}$ appears in $-\lambda_{t+1}\frac{1}{4}s_{1,t+1}$ and $+\lambda_{t+1}\nu_{1,t+1}s_{1,t+1}$ (both at β^{t+1}, exactly as for s_0 above), *and* in $-\lambda_t\nu_{0t}s_{1,t+1}$ (the $i=0$ pipeline term at date t, $s_{0t}-s_{1,t+1}$, at coefficient β^t):

$$\frac{\partial \mathcal{L}}{\partial s_{1,t+1}} = 0 \implies \beta E_t\lambda_{t+1}\nu_{1,t+1} = \tfrac{\beta}{4}E_t\lambda_{t+1} + \lambda_t\nu_{0t}. \tag{6.15}$$

Derivation: $s_{2,t+1}$

By the identical logic, one lag further along the pipeline:

$$\frac{\partial \mathcal{L}}{\partial s_{2,t+1}} = 0 \implies \beta E_t\lambda_{t+1}\nu_{2,t+1} = \tfrac{\beta}{4}E_t\lambda_{t+1} + \lambda_t\nu_{1t}. \tag{6.16}$$

PDF equation numbers: (6.14), (6.15), (6.16). Equations (6.14)–(6.16) price a project at each stage of gestation: the price of a project in its ith quarter equals the price of a project in its $(i-1)$th quarter plus 1/4 unit of goods.

Dynare (non-linear)

```
model block
exp(nu0(+1)) = 0.25;
BETTA*exp(la(+1))*exp(nu1(+1)) = (BETTA/4)*exp(la(+1)) + exp(la)*exp(
   nu0);
BETTA*exp(la(+1))*exp(nu2(+1)) = (BETTA/4)*exp(la(+1)) + exp(la)*exp(
   nu1);
```

Equation (6.14) is degenerate – λ_{t+1} cancels on both sides, leaving simply $\nu_{0,t+1} = 1/4$ for every history.

11 FOC 7: Tobin's Q – stage-3 project pricing

Derivation

$s_{3,t+1}$ is the final-quarter project that converts into installed capital via (6.7). It appears in three places: $-\lambda_{t+1}\frac{1}{4}s_{3,t+1}$ (investment aggregator, β^{t+1}); inside $\lambda_{t+1}q_{t+1}k_{t+1}\Phi(s_{3,t+1}/k_{t+1})$ (capital-accumulation term at $t+1$, β^{t+1}), whose derivative with respect to $s_{3,t+1}$ is $\lambda_{t+1}q_{t+1}\Phi'(s_{3,t+1}/k_{t+1})$ (the k_{t+1} from the outer term and the $1/k_{t+1}$ from the chain rule on Φ's argument cancel); and in $-\lambda_t\nu_{2t}s_{3,t+1}$ (the $i=2$ pipeline term at date t, β^t):

$$\frac{\partial\mathcal{L}}{\partial s_{3,t+1}} = 0 \implies \beta E_t\left[\lambda_{t+1}q_{t+1}\Phi'\left(\frac{s_{3,t+1}}{k_{t+1}}\right)\right] = \frac{\beta}{4}E_t\lambda_{t+1} + \lambda_t\nu_{2t}. \tag{6.17}$$

This links the shadow price of installed capital q_t (Tobin's Q) to the final leg of the project pipeline.
PDF equation number: (6.17).

Dynare (non-linear)

model block

```
#acp_prime_lead = 1 - PHI*(exp(s3(+1))/exp(k(+1)) - DELTA);   % Phi'(
    s3'/k')
BETTA*exp(la(+1))*exp(qq(+1))*acp_prime_lead
  = (BETTA/4)*exp(la(+1)) + exp(la)*exp(nu2);
```

12 FOC 8: capital pricing equation

Derivation

k_{t+1} appears in $-\lambda_t q_t k_{t+1}$ (capital-accumulation term at date t, coefficient β^t), and, one period later, inside $\lambda_{t+1}u_{t+1}k_{t+1}$ (capital rental income in the budget constraint at $t+1$) and inside $\lambda_{t+1}q_{t+1}\big[(1-\delta)k_{t+1} + k_{t+1}\Phi(s_{3,t+1}/k_{t+1})\big]$ (capital-accumulation term at $t+1$), both at β^{t+1}. Differentiating the latter bracket with respect to k_{t+1} via the product and chain rule,

$$\frac{\partial}{\partial k_{t+1}}\Big[k_{t+1}\Phi\big(\tfrac{s_{3,t+1}}{k_{t+1}}\big)\Big] = \Phi\big(\tfrac{s_{3,t+1}}{k_{t+1}}\big) - \frac{s_{3,t+1}}{k_{t+1}}\Phi'\big(\tfrac{s_{3,t+1}}{k_{t+1}}\big).$$

Setting the full derivative to zero and dividing by β^t:

$$\lambda_t q_t = \beta E_t\left\{\lambda_{t+1}q_{t+1}\left[1-\delta+\Phi\left(\tfrac{s_{3,t+1}}{k_{t+1}}\right) - \tfrac{s_{3,t+1}}{k_{t+1}}\Phi'\left(\tfrac{s_{3,t+1}}{k_{t+1}}\right)\right] + \lambda_{t+1}u_{t+1}\right\}. \tag{6.18}$$

This equates the price of a unit of installed capital today, q_t, to the discounted value of renting it out and then selling it, $u_{t+1}+q_{t+1}$, net of depreciation and adjustment costs.
PDF equation number: (6.18).

Dynare (non-linear)

model block

```
#x3_lead     = exp(s3(+1))/exp(k(+1));
#acp_lead    = x3_lead - (PHI/2)*(x3_lead-DELTA)^2;       % Phi(s3'/k')
#acpp_lead   = 1 - PHI*(x3_lead-DELTA);                   % Phi'(s3'/k
   ')
#u_lead      = ALFA*exp(k(+1))^(ALFA-1)*exp(h(+1))^(1-ALFA);
exp(la)*exp(qq) = BETTA*( exp(la(+1))*exp(qq(+1))*
                  (1-DELTA+acp_lead-x3_lead*acpp_lead) + exp(la(+1))*
                     u_lead );
```

13 The shadow (working-capital) interest rate

Derivation

Comparing FOC 3 (§9, eq. 6.13) to a version of the same problem without a portfolio-adjustment cost shows that the relevant rate of return on bonds is inflated by the marginal adjustment cost. Define the shadow rate

$$R_t^d \equiv \frac{R_t}{1-\Psi'(d_t)} = \frac{R_t}{1-\psi(d_t-\bar{d})}. \tag{6.19}$$

When d_t is above its steady-state level $\bar{d}$, $\Psi'(d_t) > 0$, so $R_t^d > R_t$: the shadow rate is higher than the market rate, which discourages further borrowing and pushes d_t back toward $\bar{d}$. This is exactly the R_t^d used in the firm's labor-demand condition (6.3, §2) and the household's labor-supply condition (6.12, §8).
PDF equation number: (6.19).

Dynare (non-linear)

model-local variable

```
#rd = exp(r)/(1-PSSI*(d-DBAR));
```

(already used in §2 and §8 above; repeated here for reference since it is the object being defined).

14 Trade balance and the resource identity

Derivation

Because the portfolio-adjustment cost $\Psi(d_t)$ is incurred by households, national accounts measure private consumption as $c_t + \Psi(d_t)$, not simply c_t; the trade-balance-to-GDP ratio is therefore

$$tby_t = \frac{y_t - c_t - i_t - \Psi(d_t)}{y_t}. \tag{6.21}$$

Because tby_t (and the trade balance itself, tb_t) can be close to, or cross, zero, they are kept as *level* variables rather than logged – approximating $\log(tb_t)$ around a value near zero is not meaningful.
PDF equation number: (6.21).

Dynare (non-linear)

model block

```
tb  = exp(yy) - exp(c) - exp(iv) - (PSSI/2)*(d-DBAR)^2;
tby = tb/exp(yy);
```

(`tb` and `tby` are declared as plain, non-logged variables in the `var` block, unlike every other quantity variable.)

15 Driving forces: the estimated interest-rate processes

Derivation

The model is closed with the (limited-information, VAR-based) estimated law of motion for the country interest rate, in log-deviations from steady state (denoted $\widehat{\cdot}$):

$$\widehat{R}_t = 0.635\,\widehat{R}_{t-1} + 0.501\,\widehat{R}^{us}_t + 0.355\,\widehat{R}^{us}_{t-1} - 0.791\,\widehat{y}_t + 0.617\,\widehat{y}_{t-1} + 0.114\,\widehat{iv}_t - 0.122\,\widehat{iv}_{t-1} + 0.288\,tby_t - 0.190\,tby_{t-1} + \epsilon^r_t, \tag{6.20}$$

where ϵ^r_t is an i.i.d. disturbance with mean zero and standard deviation 0.031. Because (6.20) involves the exogenous world interest rate, its own law of motion must also be part of the equilibrium system:

$$\widehat{R}^{us}_t = 0.830\,\widehat{R}^{us}_{t-1} + \epsilon^{rus}_t. \tag{6.22}$$

Both ϵ^r_t and ϵ^{rus}_t are the two structural shocks driving the model; the text's own Figure 6 traces out the theoretical impulse responses to each in turn.
PDF equation numbers: (6.20), (6.22).

Dynare (non-linear)

model block

```
% eq. (6.22) -- world interest rate
rus - log(RUSSS) = ARUSRUS1*(rus(-1)-log(RUSSS)) + eps_rus;

% eq. (6.20) -- country interest rate
r - log(RSS) = ARRUS*(rus-log(RUSSS))     + ARRUS1*(rus(-1)-log(RUSSS
    ))
             + ARY*(yy-log(YYSS))          + ARY1*(yy(-1)-log(YYSS))
             + ARIV*(iv-log(IVSS))         + ARIV1*(iv(-1)-log(IVSS))
             + ARTBY*(tby-TBYSS)           + ARTBY1*(tby(-1)-TBYSS)
             + ARR1*(r(-1)-log(RSS))       + eps_r;
```

(`RSS`, `RUSSS`, `YYSS`, `IVSS`, `TBYSS` denote the steady-state levels of R, R^{us}, y, i, tby – see the assembled model in §14 – and `eps_r`, `eps_rus` are declared as `varexo` innovations directly, which is the natural Dynare counterpart of the i.i.d. shocks $\epsilon^r_t, \epsilon^{rus}_t$ described in the text.)